Assignment 1

Submitted by: Kirandeep Kaur, Roll no 1024170280, Group 3Q25

1. Write Java Program to Check Leap Year

```java
import java.util.*;
public class LeapYear {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a year: ");
        int year = sc.nextInt();
        if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0)) {
            System.out.println(year + " is a Leap Year.");
        } else {
            System.out.println(year + " is Not a Leap Year.");
        }
    }
}
```

Output:

```
Enter a year: 2024
2024 is a Leap Year.
```

```
Enter a year: 2026
2026 is Not a Leap Year.
```

2. Write Java Program to Check Whether a Character is Alphabet or Not

```java
import java.util.*;
class CheckAlphabet {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter a character: ");
        char ch = sc.next().charAt(0);
        if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z')) {
            System.out.println(ch + " is an alphabet.");
        }
        else {
            System.out.println(ch + " is not an alphabet.");
        }
    }
}
```

Output

```
Enter a character:
5
5 is not an alphabet.
```

```
Enter a character:
k
k is an alphabet.
```

3. Write Java Program to Find Factorial of a Number

```java
import java.util.*;
class Factorial {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();
        long fact = 1;
        for (int i = 1; i <= n; i++) {
            fact *= i;
        }
        System.out.println("Factorial of " + n + " is " + fact);
    }
}
```

Output

```
Enter a number: 5
Factorial of 5 is 120
```

4. Write Java Program to Display Fibonacci Series

```java
import java.util.*;
public class Fibonacci{
    public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      int n,a,b,c;
      System.out.print("Enter the number of terms: ");
      n = sc.nextInt();
      a = 0;
      b = 1;
      System.out.println("Fibonacci Series: ");
      for(int i = 1; i <= n; i++){
        System.out.println(a+" \t");
        c = a + b;
        a = b;
        b = c;
      }
    }
}
```

Output

```
Enter the number of terms: 8
Fibonacci Series:
0
1
1
2
3
5
8
13
```

5. Write Java Program to Find GCD of two Numbers

```java
import java.util.*;
public class Gcd{
    static int findgcd(int a, int b){
      return b == 0 ? a : findgcd(b, a%b);
    }
    public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      int a,b;
      System.out.println("Enter 2 numbers");
      a = sc.nextInt();
      b = sc.nextInt();
      int ans = findgcd(a,b);
      System.out.println("GCD of " + a + " and " + b + " is " +ans);
    }
}
```

Output

```
Enter 2 numbers
20
28
GCD of 20 and 28 is 4
```

6. Write Java Program to Find LCM of two Numbers

```java
import java.util.*;
public class Lcm{
    static int findgcd(int a, int b){
      return b == 0 ? a : findgcd(b, a%b);
    }

    static int findlcm(int a, int b){
      return (a/findgcd(a,b)) * b;
    }
    public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      int a,b;
      System.out.println("Enter 2 numbers");
      a = sc.nextInt();
      b = sc.nextInt();
      int ans = findlcm(a,b);
      System.out.println("LCM of " + a + " and " + b + " is " +ans);
    }
}
```

Output

```
Enter 2 numbers
120
72
LCM of 120 and 72 is 360
```

7. Write Java Program to Count Number of Digits in an Integer

```java
import java.util.*;
public class CountDigits {
    public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      int n,t,c;
      System.out.println("Enter n");
      n = sc.nextInt();
      t = n;
      c = 0;
      while(t>0){
        c++;
        t /= 10;
      }
      System.out.println(n+" has " +c+ " digits");
    }
}
```

Output

```
Enter n
1234
1234 has 4 digits
```

8. Write Java Program to Reverse a Number

```java
import java.util.*;
public class Reverse{
    public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      int n,t,s;
      System.out.println("Enter n");
      n = sc.nextInt();
      s = 0;
      while(n>0){
        int r = n%10;
        s = s*10 + r;
        n /= 10;
      }
      System.out.println("Reversed number : " +s);
    }
}
```

Output

```
Enter n
453
Reversed number : 354
```

9. Write Java Program to Calculate the Power of a Number

```java
import java.util.*;
public class Power{
    public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      int n,x,a;
      System.out.println("Enter the number and its power");
      x = sc.nextInt();
      n = sc.nextInt();
      a = 1;
      for(int i=1; i<=n; i++){
        a = a*x;
      }
      System.out.println("Result : " +a);
    }
}
```

Output

```
Enter the number and its power
2
5
Result : 32
```

10. Write Java Program to Check Palindrome

```java
import java.util.*;
public class Palindrome{
    public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      int n,t,s;
      System.out.println("Enter n");
      n = sc.nextInt();
      t = n;
      s = 0;
      while(t>0){
        int r = t%10;
        s = s*10 + r;
        t /= 10;
      }
      if (s == n)
            System.out.println(n + " is a Palindrome.");
      else
            System.out.println(n + " is Not a Palindrome.");
    }
}
```

Output

```
Enter n
478
478 is Not a Palindrome.
```

```
Enter n
737
737 is a Palindrome.
```